Problem 9.18

A deposit is made on January 1, 2004. The investment earns 6% compounded semi-annually. Calculate the monthly effective interest rate for the month of December 2004.

Solution

The nominal interest rate compounded semi-annually is

$$i^{(2)} = 6\%.$$

Therefore, the effective interest rate for each six-month period is

$$\frac{i^{(2)}}{2} = \frac{0.06}{2} = 0.03.$$

Let j be the monthly effective interest rate. Since six months at rate j must be equivalent to one semi-annual period at rate 0.03, we have

$$(1+j)^6 = 1.03.$$

Taking the sixth root of both sides,

$$1 + j = (1.03)^{1/6}.$$

Thus,

$$j = (1.03)^{1/6} - 1.$$

Finally,

$$j \approx 0.004938622.$$

Hence, the monthly effective interest rate for December 2004 is

$$0.4938622\%.$$